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REQUEST PROCESSING SYSTEM

Abstract

There is provided a request processing system for receiving a request sent from the first apparatus at the second apparatus and executing the request, wherein the first apparatus comprises: means for writing the generated first apparatus-identifying specific data into a predetermined address on a communication network; and means for sending a request with a signature, signed with the secret key, and the public key to the second apparatus, wherein the second apparatus comprises: means for reading the first apparatus-identifying specific data; means for authenticating the first apparatus by generating, from the public key, public key data enable to uniquely identify the public key; and determining whether this public key is included in the downloaded first apparatus-identifying specific data as well as, based on the determination, validating a signature of the request with a signature based on the public key received from the first apparatus and the unlocking cryptographic data included in the downloaded first apparatus-identifying specific data; and means for executing the validated request based on the authentication of the first apparatus.

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Background/Summary

FIELD OF THE INVENTION
[0001] The present invention is related to a request processing system and a method thereof for processing a request to lock or unlock a vehicle, house, locker, or the like using an electronic device such as a smart key.
BACKGROUND OF THE INVENTION
[0002] Practical applications have been developed for a smart key system for locking or unlocking a vehicle, a locker, or the like without using a physical key, and the following literatures are listed as prior art.
Prior Art Document
Patent Document
Patent Document 1
[0003] Japanese Unexamined Patent Application Publication No. 2006-9333
Patent Document 2
[0004] Japanese Unexamined Patent Application Publication No. 2009-275363
Non-Patent Document
SUMMARY OF THE INVENTION
Problem to be Solved by the Invention
[0005] Here, according to certain statistics, an average number of years of use of an automobile in Japan as of 2017 was 12.91 years. Meanwhile, as for smart keys for locking and unlocking vehicles, certain private corporations manage their respective systems in general, and there is no guarantee that those corporations can continuously provide their managing services over the entire years of automobile use. Also, excessively high managing expenses of the smart keys makes it difficult to continue those services due to overbearing burden.
[0006] Considering the above situation, the purpose of the present invention is to provide a system capable of providing a service more universally without relying on a particular service provider in a request processing apparatus such as ones processed between a smart key and a vehicle onboard apparatus.
Means for Solving the Problem
[0007] In order to achieve the above object, the present inventors considered-in utilizing a technology for authenticating a first apparatus and a second apparatus using a public key-that storing an actual public key in a server will overload the server's storage area, devised an apparatus capable of accurate authentication without sharing the public key on a network, and validated that concept in good faith to finally complete the present invention.
[0008] In other words, according to a principal aspect of the present invention, the following invention is provided.
[0009] A request processing system, comprising: [0010] (1) a first apparatus and a second apparatus associated with the first apparatus, [0011] the request processing system being for receiving a request sent from the first apparatus at the second apparatus and executing the request,

[0012] wherein the first apparatus comprises: [0013] means for maintaining a specific secret key and a public key; [0014] means for generating first apparatus-identifying specific data, including public key data enable to uniquely identify the public key and smaller in size than the public key, and unlocking cryptographic data enable to uniquely identify unlocking cryptographic information; [0015] means for writing the generated first apparatus-identifying specific data into a predetermined address on a communication network; and [0016] means for sending a request with a signature, signed with the secret key, and the public key to the second apparatus, [0017] wherein the second apparatus comprises: [0018] means for reading the first apparatus-identifying specific data written at the predetermined address; [0019] means for authenticating the first apparatus by generating, from the public key received from the first apparatus, public key data enable to uniquely identify the public key; and determining whether this public key is included in the downloaded first apparatus-identifying specific data as well as, based on the determination, validating a signature of the request with a signature based on the public key received from the first apparatus and the unlocking cryptographic data included in the downloaded first apparatus-identifying specific data; and [0020] means for executing the validated request based on the authentication of the first apparatus.

[0021] According to such a configuration, when a request is sent from the first apparatus to the second apparatus, the public key is sent to the second apparatus together with this request. Further, at this second apparatus the above public key may be uniquely identified; the smaller-sized public key is generated for comparison with the specific data retrieved from the predetermined address on the network; and a particular unlocking data is used to validate the signature added to the request to thereby authenticate the user terminal.

[0022] In this manner, since the actual public key may be directly sent from the first apparatus to the second apparatus, the shared information does not need to be the public key itself, and the “public key data,” which is smaller in size than the public key, may be placed at a predetermined address on the network. Thus, an effect may be achieved in that a server storage area for managing the smart key and the like are not overloaded.

[0023] Here, according to one embodiment of the present invention, it is preferred that the first apparatus is a smart key, and that the second apparatus is mounted on a vehicle and is a vehicle onboard apparatus for controlling unlocking and locking of the vehicle by the request of the first apparatus.

[0024] Also, according to another embodiment of the present invention, the public key data is preferably generated by using a hash function to convert the public key; and more preferably, the last 20 bytes extracted of this hash value as well as a public key address used in a blockchain system.

[0025] According to yet another embodiment of the present invention, the first apparatus-identifying specific data is preferably written in a smart contract address of Ethereum. Thus, storing the user terminal-specific identification code on a blockchain network ensures the universality of code storage.

[0026] According to still another embodiment of the present invention, this system preferably sends the first apparatus-identifying information to the predetermined address in order to authenticate the writing authority by the first apparatus.

[0027] The above and other characteristics of the present invention will be readily appreciated by those skilled in the art by referring to the following Detailed Description of the Invention and the accompanying drawings.

Description

Effect of the Invention

BRIEF DESCRIPTION OF THE DRAWINGS

[0028] FIG. **1** is an overall configuration diagram showing a system according to one embodiment of the present invention;

[0029] FIG. **2** is a system configuration diagram showing a smart key according to the one embodiment;

[0030] FIG. **3** is a schematic structural view showing a vehicle onboard apparatus according to the one embodiment;

[0031] FIG. **4** is a schematic structural view showing a control section of the vehicle onboard apparatus according to the one embodiment;

[0032] FIG. **5** is a system configuration diagram showing a smart key management server according to the one embodiment;

[0033] FIG. **6** is a flowchart showing the steps of setting a smart contract address according to the one embodiment;

[0034] FIG. **7** is a flowchart showing the steps of issuing a user terminal-specific identification code according to the one embodiment;

[0035] FIG. **8** is a schematic view showing a user interface of a smart key according to the one embodiment;

[0036] FIG. **9** is a explanatory view describing a structure of the user terminal-specific identification code according to the one embodiment;

[0037] FIG. **10** is a flowchart showing the steps of sending a request according to the one embodiment; and

[0038] FIG. **11** is a flowchart showing the steps of processing a request according to the one embodiment.

DETAILED DESCRIPTION OF THE INVENTION

[0039] One embodiment of the present invention will be described below in accordance with accompanying drawings.

[0040] FIG. **1** is an overall configuration diagram showing an example of applying a request processing system of the present invention to a smart key system **1** for performing locking and unlocking of a vehicle door.

Overall Configuration

[0041] This smart key system **1** comprises one or multiple user terminals **2** (corresponding to a “first apparatus” of the present invention); a smart contract **4** (a “predetermined address on a network” of the present invention) connected with this user terminal **2** via the Internet **3** (corresponding to the “network” of the present invention); a vehicle onboard apparatus **5** (corresponding to a “second apparatus” of the present invention) mounted on a vehicle; and a smart key management server **6** for managing this smart key system **1**.

[0042] In this smart key system **1**, the user terminal **2** writes, in the smart contract **4** in advance, a user terminal-specific identification code (specific data for identifying the first apparatus) **7** (A). On the other hand, the vehicle onboard apparatus **5** retrieves the user terminal-specific identification code **7** from the smart contract **4** (B) and stores it in the user terminal **2**. Then, if a user of the vehicle uses the user's own user terminal **2** to send a particular request to the vehicle onboard apparatus **5** (C), the vehicle onboard apparatus **5** is configure to use the user terminal-specific identification code **7** to perform qualification authentication of the user terminal **2**.

Smart Contract

[0043] Here, the smart contract **4** is a computer protocol built on the Ethereum, and is configured such that a transaction is automatically performed if a contract and its execution condition are programmed in advance and if that contract condition is met.

[0044] In Ethereum, it is configured so that an execution history of a smart contract **4** is recorded in a blockchain on a P2P network called Ethereum network. Ethereum has a tunable programming language for describing a smart contract **4**, and a network participant is enabled to describe and execute an arbitrary smart contract **4** in a blockchain on this network. Such a mechanism has enabled executing a program and sharing its result with the entire P2P as an execution environment without depending on a particular central managing organization.

User Terminal

[0045] The user terminal **2** is typically a smartphone, but it may be a tablet or any other computer device.

[0046] Also, in this embodiment, there are configured three types of users using this user terminal **2**: an owner possessing a vehicle, the owner's family, and a car-sharing user. For example, in a car-sharing service, an operator of a car-sharing business is an owner, whereas a person sharing a vehicle is a car-sharing user merely utilizing a vehicle. Similarly for a rent-a-car, an operator of a rent-a-car business is an owner possessing vehicles, and a person who rents a vehicle is a car-sharing user utilizing the vehicle. Also, in case of a private car, a registered owner of the car is an owner, and the owner's family member driving the car is a user utilizing the vehicle.

[0047] In this embodiment, as discussed in detail below, there are different authorities for writing into a smart contract **4** according to the user type. cl Vehicle Onboard Apparatus

[0048] The vehicle onboard apparatus **5** is an electronic apparatus mounted on a vehicle, and equipped with a function for controlling locking and unlocking of a door of the vehicle. In addition to the above function, this vehicle onboard apparatus **5** may be equipped with other functions, for example, a highway toll payment function, a navigation audio visual function, and the like.

[0049] In this embodiment, each vehicle onboard apparatus **5** is assigned with a specific address (smart contract address) of the smart contract **4**, and the user terminal **2** writes the above user terminal-specific identification code **7** in this smart contract address (indicated with A in FIG. **1**). Then, the vehicle onboard apparatus **5** retrieves and stores the written user terminal-specific identification code **7** (indicated with B in FIG. **1**). Subsequently, upon receiving a request with a signature, and a public key from the user terminal **2** (indicated with C in FIG. **1**), the vehicle onboard apparatus **5** compares the above user terminal-specific identification code **7** and the public key, and also validates the signature of the request with a signature (for unlocking or locking) to thereby authenticate the user terminal **2**, and then, process the request as needed.

Smart Key Management Server

[0050] Also, the smart key management server **6** is, for example, operated by the present applicant, and it has a function for generating a smart contract **4** on a blockchain for each vehicle onboard apparatus as well as a function for setting an address of the smart contract in the user terminal **2** and the vehicle onboard apparatus **5**.

[0051] Configuration of each of the devices **2-6** will be described in detail below.

(Detail Configuration of User Terminal

[0052] FIG. **2** is a schematic structural view showing the user terminal **2**.

[0053] The user terminal **2** is defined by a CPU **10**, a RAM **11**, and an input/output section **12**, which are connected with a bus **13**, which is in turn connected with a data storage section **14** and a program storage section **15**. In this embodiment, connected to the input/output section **12** are a request input device **16** such as a button, a keyboard, a touchscreen, and/or the like for entering a request to the vehicle onboard apparatus **5**; a network communication device **17** for communicating the network **3**; and a onboard apparatus wireless communication device **18**.

Data Storage Section

[0054] The data storage section **14** stores therein—to only list data related to the present invention —user secret key/public key **20**; a user ID **21** (Ethereum address in this embodiment); a user type **22** (owner, family, or share); a user key attribute **23** (key expiration date, etc.); unlocking cryptographic information **24**; the user terminal-specific identification code **7**; request command information **26** (unlocking, locking, etc.); and a smart contract address **27** (Ethereum address).

Program Storage Section

[0055] Also, the program storage section **15** stores therein—to only list configurations related to the present invention—a user secret key/public key generation section **28**; a user authentication section **29**; a public key address generation section **30**; a user terminal-specific identification code generation section **31**; a specific identification code transmitting section **32**; a request receiving section **33**; a request-with-a-signature generation section **34**; and a request-with-a-signature/public key transmitting section **35**.

[0056] Note that the data storage section **14** and the program storage section **15** are namely auxiliary storage devices (HDD and/or SDD) of this user terminal **2**. Also, the respective functional sections **29-35** stored in the above program storage section **15** are configured as a “smart key app **19**,” in which the respective functional sections **29-35** are partially or fully installed in the user terminal **2** (see FIG. **1**). Moreover, when this app **19** is installed, space is secured for storing the respective information **20-27** of the data storage section **14** in the auxiliary storage device of the user terminal **2**.

[0057] Note that the respective information **20-27** of this data storage section **14** may be respective tables of a database and their values stored therein, or may be directly written as in programs in the respective functional sections **29-35** stored in the program storage section **15**. Also, since the user's secret key needs to be managed in the safest-possible way, the user secret key is preferably stored in a secure element provided by the user terminal (smartphone, etc.).

[0058] Further, the respective functional sections **29-35** stored in the program storage section **15** are adapted to be called, deployed, and executed as needed on the RAM **11** by the CPU **10** to thereby function as respective components described in the claims of the present application.

Vehicle Onboard Apparatus

[0059] FIG. **3** is a schematic structural view showing the vehicle onboard apparatus **5**.

[0060] This vehicle onboard apparatus **5** has a control section **41**, a transmitting/receiving section **42** connected with this control section **41**, a operating section **43**, a display section **44**, and an audio output section **45**. The above transmitting section **42** is provided with an antenna **40** for directly and wirelessly communicating with the user terminal **2**, and also adapted to be capable of connecting with the Internet **3** in order to communicate with the smart contract **4** and the smart key management server **6**.

[0061] FIG. **4** is a function block diagram showing a configuration implemented in the control section **41**.

[0062] This control section **41** has a user terminal-specific identification code receiving section **47**, a request-with-a-signature/public key receiving section **48**, a public key address generation section **49**, a user terminal authentication section **50**, and a request processing section **51**. Also, this control section **41** is adapted such that it may store the smart contract address **27** and the user terminal-specific identification code **7** in a memory.

Smart Key Management Server Further, FIG. **5** is a function block diagram showing the simplified smart key management server **6**.

[0063] To only show configurations related to the present invention, this smart key management server **6** has a vehicle/vehicle onboard apparatus control section **53**, a smart contract issuing section **54**, a smart contract address storage section **55**, and a user ID receiving section **56**.

[0064] The above configuration will be discussed in detail below in reference with their respective operations.

Installation and Activation of Smart Key App

[0065] Firstly, when the above user terminal is a smartphone or the like, the system shown in FIG. **2** is configured by installing and activating the smart key app **19** described above. In this embodiment, during this installation and activation, a user ID is set to this smart key app **19**, and based on this user ID, the user secret key/public key generation section **28** generates a user secret key/public key **20** and stores it in the data storage section **14**.

[0066] Note that these user secret key and public key may be generated using an OS of the user terminal, or may be generated by another apparatus (for example, the smart key management server **6** or the like) for use.

Generation of Smart Contract

[0067] FIG. **6** is a flowchart showing steps of issuing a smart contract at the smart key management server **6**.

[0068] Firstly, the user ID receiving section **56** receives a user ID **21** from the user terminal **2** (Step S**1**).

[0069] Next, the smart contract issuing section **54** refers to information of a vehicle onboard apparatus stored in the vehicle/vehicle onboard apparatus management section **53**, and issues a smart contract for each vehicle/vehicle onboard apparatus (Step S**2**). Here, the previously received user ID is specified as "user type: owner" via constructor parameters of the smart contract **4**.

[0070] After the smart contract **4** is issued, the Ethereum network generates a smart contract address (Ethereum address) **27**, which will be stored in the smart contract address storage section **55** in association with the vehicle/vehicle onboard apparatus (Step S**3**). This generated address will be a unique identifier of the smart contract.

[0071] Subsequently, this smart contract address **27** is registered in a smartphone app **19** (smart key) on the user terminal **2** of the vehicle owner, and a vehicle onboard apparatus **5** (Step S**4**) (see FIG. **2**, FIG. **4**). Specific registration method varies depending on the system, but it may be done online or installed offline.

[0072] Thus, the vehicle onboard apparatus **5** and the user terminal **2** will share the same smart contract address **27**, and therefore, it will be possible to write data in a particular smart contract **4**, refer to the user ID **21**, and with authority, write data into the smart contract **4**.

[0073] Note that, in this embodiment, each individual user has an Ethereum address and uses it as the user ID.

User Authentication at User Terminal

[0074] FIG. **7** is a flowchart showing operations on the user terminal **2**.

[0075] Firstly, the user authentication section **29** performs user authentication on this user terminal **2** (Step S**5**).

[0076] In this embodiment, user authentication utilizes a password and/or biometrics authentication executed by the OS of this terminal **2**. Note that it may be configured such that the user has a member account in the smart key management server **6**, wherein a predetermined ID and a password may be used to log into the member account.

[0077] Upon successful authentication by this user authentication section **29**, this smart key app **19** displays a user interface **58**, as shown in FIG. **8** (*a*), on the user terminal **2**.

Generation of Public Key Address by User Terminal

[0078] When the user presses a code initialization button (corresponding with the above-mentioned request input device), indicated with **60** in the figure, on the user interface **58**, the public key address generation section **30** and the specific identification code generation section **31** refers to the user secret key/public key **20** to generate a public key address sized smaller than the number of bytes of the public key, and utilizes this public key address to generate the user terminal-specific identification code **7** (Steps S**6**-S**10**).

[0079] These operations of the public key address generation section **30** and the specific identification code generation section **31** will be described below in an example where a public key cryptographic algorithm is the ECDSA.

[0080] Firstly, the public key address generation section **30** retrieves the public key (Step S**6**). An ECDSA public key is 65-byte long.

[0081] Next, the public key address generation section **30** hashes the ECDSA public key using a hash function (Keccak-256) to generate a 256-bit (32-byte) hush value (Step S**7**). Then, it extracts the latter 20 bytes from the 32-byte hash value as a "public key address" (Step S**8**).

[0082] In this embodiment, one reason for changing the public key address data size down to 20 byte is because that has been validated to be one of the minimum sizes in order to identify the public key uniquely enough. Therefore, as long as this purpose is attainable, the public key address may even have a smaller number of bytes than 20, and of course, a greater number of bytes than 20.

Generation of Identification Specific Code by User Terminal

[0083] When the public key address is generated as above, the user terminal-specific identification code generation section **31** couples this public key address with unlocking cryptographic information **24** (a unique identifier of a signature scheme and a parameter of a signature scheme) and stores it as a user terminal-specific identification code **7**, which is one value (Steps S**9**, S**10**).

[0084] Specifically, as shown in FIG. **9**, public key address **62** (20 bytes) is coupled with the unlocking information (a unique identifier of a signature scheme and a unique identifier of a parameter of a signature scheme) (for ECC, “1.2.840.10045.2.1” and “1.3.132.0.10,” respectively) into one value, which will be binary-encoded to generate an individual identification code value to be stored. As a specific data format, in this embodiment, the SubjectPublicKeyInfo (SPKI) format of ASN.1 DER-encoded X.509 is used.

[0085] In such a configuration as the above, the smaller the size of an actual public key is, the smaller the size of the entire SPKI-specific identification code **7** also becomes, allowing to reduce a storage area in the smart contract, for example, from 3 slots to 2 slots (the present embodiment).

[0086] Note that, typically, in the public key data, there is no unlocking cryptographic information (public key cryptographic method and parameters). Therefore, only sharing the public key is meaningless; one would not know how to use the public key unless its context or cryptographic information is also known.

[0087] Note that, when sharing the public key of an Ethereum user account, only the actual public key data may be shared; this is because public key cryptographic information (method: ECC, parameter: secp256k1) is fixed by context (Ethereum specifications).

[0088] Whereas, in the smart key system according to the embodiment of the present invention, a signing method for the request with a signature, discussed below, needs to accommodate a plurality of public key cryptography methods, and (or) the context does not limit the public key cryptographic information. When sharing a public key in such an environment, public key cryptographic information also needs to be shared.

[0089] Also, in theory, it is possible to share the public key address **62** and unlocking cryptographic information **63** separately. However, since that would be inconvenient, in this example, public key cryptographic information and actual public key data are generated and shared as one value (code) (see FIG. **9**).

Writing Specific Identification Code by User Terminal

[0090] Next, the specific identification code transmitting section **32** refers to the smart contract address **27** and writes the specific identification code **7** generated as above in the smart contract **4** (Step S**11**).

[0091] Also, in this embodiment, the specific identification code transmitting section **32** writes the following information in addition to the above codes (Step S**11**). [0092] User ID (in this embodiment, Ethereum address of the user is used as the user ID) [0093] User type (owner, family, share, etc.) [0094] User's key attribute (e.g., effective period of the key; vehicle speed limit when this key was used for unlocking, etc.; uniquely formatted binary data encoded with DER).

[0095] By writing the four values including the above additional values in the smart contract **4**, specification on unlocking a vehicle will become possible by whom (user address), with what authority (user type), with which key (user's key), and during what period of time (user key attribute).

[0096] Note that, in this embodiment, one smart contract **4** is adapted to be capable of storing therein “user information” of one vehicle owner (type: owner) and “user information” of a plurality of users (type: family, share, etc.).

[0097] Also, only a user having a user ID of user type "owner" may write into a smart contract.
[0098] Note that, for the user authority authentication method, this embodiment is adapted to use a conventional authentication method of Ethereum.
[0099] Specifically, the authentication is executed with the following steps. [0100] (a) The Ethereum network recovers a public key from a transaction signature and transaction data. [0101] (b) The Ethereum network generates an Ethereum address (user ID) of a user from the recovered public key. [0102] (c) The Ethereum network passes over the generated Ethereum address to a smart contract as a sender address of the transaction. [0103] (d) Finally, on a vehicle smart contract side, the sender address of the transaction delivered to the smart contract is checked, wherein a writing transaction will be permitted if it matches the user ID (Ethereum address) of the user type: owner, and the transaction will not be permitted if it does not match.
[0104] Note that, in this embodiment, this code generation is done by the app **19** installed in the user terminal **2**, but the external smart key management server **6** may perform the encoding, and it may even perform the writing into the smart contract **4** as described herein below.
Sending Request With Signature by User Terminal
[0105] Next, request transmission operations by the user terminal **2** will be described with respect to a flowchart shown in FIG. **10**.
[0106] Firstly, after a specific identification code **7** is written in a smart contract through the interface of FIG. **8** (*a*), by pressing the main screen button **65**, a request command input interface will be displayed, as indicated with **64** in FIG. **8** (*b*).
[0107] In this example, as input-permitted requests, unlock **66**, lock **67**, start engine **68**, and open trunk **69** are displayed. However, the present invention is not so limited.
[0108] Subsequently, when the user presses any of the buttons, that request is received (Step S**12**), and request command information **26** corresponding with that button will be retrieved from the data storage section **14** followed by a request with a signature being generated by the request-with-a-signature generation section **34** (Step S**13**). Specifically, an electronic signature is created by using a secret key in user secret key/public key information to encode the request, and a request with a signature including that electronic signature is generated.
[0109] Next, the request-with-a-signature transmitting section **35** sends a request with a signature and a public key for utilizing a vehicle to the vehicle onboard apparatus **5** (Step S**14**). This request may include a vehicle usage condition (e.g., date/time of use) in addition to the request itself. The request with a signature and the public key are sent to the vehicle onboard apparatus **5** wirelessly via the wireless communication device.
[0110] Note that the transmission method is not limited to wireless communication and, for example, transmission may be performed via a network using a public wireless network.
Operation of Vehicle Onboard Apparatus
[0111] Next, operations of the vehicle onboard apparatus **5** will be described with respect to a flowchart of FIG. **11**.
[0112] Firstly, as discussed above, it is presumed that the smart contract address **27** issued by the smart key management server **6** has been set in this vehicle onboard apparatus **5**. This address is set offline or online in the vehicle onboard apparatus **5** as previously mentioned.
Retrieving Identification Code From Smart Contract at Vehicle Onboard Apparatus
[0113] Next, the user terminal-specific identification code receiving section **47** refers to the smart contract address **27** and retrieves the user terminal-specific identification code **7** from a smart contract according to the address **27** via the transmitting/receiving section **42**.
[0114] Timing to download this specific identification code **7** is arbitrary, but for example, the specific identification code **7** may be requested on a regular basis or according to a predetermined schedule.
Receiving Request With a Signature From User Terminal at Vehicle Onboard Apparatus
[0115] The request-with-a-signature receiving section **48** receives a request with a signature and a

public key sent from the user terminal **2** through the transmitting/receiving section **42** (Step S**15**).

Generation of Public Key Address at Vehicle Onboard Apparatus

[0116] Subsequently, the public key address generation section **49** a generates a public key address from the public key received by the request-with-a-signature receiving section **48**. This generation is executed in a similar method to the generation of the public key address **62** by the public key address generation section **30** of the user terminal as discussed above (Steps S**16**, S**17**).

Authentication of User Terminal at Vehicle Onboard Apparatus

[0117] Next, the user terminal authentication section **50** uses the public key address generated as above to authenticate the user terminal **2**. This authentication process is performed through the next two steps (Steps S**18**, S**19**).

[0118] Firstly, the user terminal authentication section **50** searches for the public key address generated as above in the identification code **7** retrieved from the smart contract (FIG. **9**), and if a matching code character string (public key address **62**) exists, it is tentatively determined that the public key and the request with a signature have been sent from a predetermined user terminal **2** (Step S**18**).

[0119] However, this alone is not enough to authenticate the user terminal. It is because both of the smart contract content and the public key are public information.

[0120] Thus, the user terminal authentication section **50** validates the request with a signature using the public key (Step S**19**). If this validation yields a positive result, the request with a signature is authenticated to have been sent from a predetermined user terminal **2**.

[0121] Next, the request processing section **51** receives the above authentication result to execute processing based on the validated request (Step S**20**).

[0122] For example, in case of vehicle onboarding, if an unlocking request is sent from the user terminal **2** to the vehicle onboard apparatus **5**, it unlocks the vehicle in response. Similarly, if a request including locking is sent, the vehicle onboard apparatus **5** locks the vehicle in response.

[0123] According to the configuration discussed above, since the actual public key may be directly sent from the user terminal **2**, which is the first apparatus, to the vehicle onboard apparatus **5**, which is the second apparatus, the information shared on blockchain does not need to be the public key itself, and the “public key data,” which is smaller in size than the public key, may be placed at a predetermined address on the network. Accordingly, an effect may be achieved in that a server storage area for managing the smart key and the like are not overloaded.

[0124] It should be mentioned that the present invention is not limited by the above embodiment, and that various changes and modifications can be made without departing from the scope and spirit of the present invention.

[0125] For example, the above embodiment example illustrated an electronic key system utilizing a blockchain, but this is just an example, and a server on network may be utilized in place of the blockchain.

[0126] Also, in the above one embodiment example, a smart key for unlocking and/or locking a vehicle was described as an example, but the present invention is not limited to this example, and may be applied to any system required to authenticate a first apparatus when sending a request from the first apparatus to a second apparatus.

[0127] For instance, the present invention may be applied to a smart key used for a home door, a hotel room key, a home delivery locker, or a smart device for launching a particular device such as a computer, wherein any type of the request may be used.

[0128] Moreover, in the above one embodiment, the secret key and the public key were generated when the smart key app was installed, but the present invention is not so limited. For example, they may be generated when the smart key code initialization button **60**, shown in FIG. **8**, is pressed.

DESCRIPTION OF THE REFERENCE NUMBERS

[0129] ID. User [0130] **1**. Smart key system [0131] **2**. User terminal [0132] **3**. Internet [0133] **4**. Smart contract [0134] **5**. Vehicle onboard apparatus [0135] **6**. Smart key management server [0136]

7. User terminal-specific identification code [0137] **10**. CPU [0138] **11**. RAM [0139] **12**. Input/output section [0140] **13**. Bus [0141] **14**. Data storage section [0142] **15**. Program storage section [0143] **16**. Request input device [0144] **17**. Network communication device [0145] **18**. Onboard apparatus wireless communication device [0146] **19**. Smart key app [0147] **20**. User secret key/public key [0148] **21**. User ID [0149] **22**. User type [0150] **23**. User key attribute [0151] **24**. Unlocking cryptographic information [0152] **26**. Request command information [0153] **27**. Smart contract address [0154] **29**. User authentication section [0155] **30**. Public key address generation section [0156] **31**. Specific identification code generation section [0157] **32**. Specific identification code transmitting section [0158] **33**. Request receiving section [0159] **34**. Request-with-a-signature generation section [0160] **35**. Request-with-a-signature/public key transmitting section [0161] **40**. Antenna [0162] **41**. Control section [0163] **42**. Transmitting/receiving section [0164] **43**. Operating section [0165] **44**. Display section [0166] Audio output section [0167] **47**. User terminal-specific identification code receiving section [0168] **48**. Request-with-a-signature/public key receiving section [0169] **49**. Public key address generation section [0170] **50**. User terminal authentication section [0171] **51**. Request processing section [0172] **53**. Onboard apparatus management section [0173] **54**. Smart contract issuing section [0174] **55**. Smart contract address storage section [0175] **56**. User ID receiving section [0176] **58**. User interface [0177] **62**. Public key address [0178] **63**. Unlocking cryptographic information [0179] **65**. Main screen button [0180] **66**. Unlock button [0181] **67**. Lock button [0182] **68**. Start engine button [0183] **69**. Open trunk button

Claims

1. A request processing system having a first apparatus and a second apparatus associated with the first apparatus for receiving a request sent from the first apparatus at the second apparatus and executing the request, wherein the first apparatus comprises: a unit for maintaining a specific secret key and a public key; a unit for generating first apparatus-identifying specific data, including public key data enable to uniquely identify the public key and unlocking cryptographic data enable to uniquely identify unlocking cryptographic information; a unit for writing the generated first apparatus-identifying specific data into a predetermined address on a communication network; and a unit for sending a request with a signature, signed with the secret key, and the public key to the second apparatus, wherein the second apparatus comprises: a unit for reading the first apparatus-identifying specific data written at the predetermined address; a unit for authenticating the first apparatus by generating, from the public key received from the first apparatus, public key data enable to uniquely identify the public key; and determining whether this public key is included in the downloaded first apparatus-identifying specific data as well as, based on the determination, validating a signature of the request with a signature based on the public key received from the first apparatus and the unlocking cryptographic data included in the downloaded first apparatus-identifying specific data; and a unit for executing the validated request based on the authentication of the first apparatus.
2. The request processing system of claim 1, wherein the first apparatus is a smart key, and the second apparatus is mounted on a vehicle and is a vehicle onboard apparatus for controlling unlocking and locking of the vehicle by the request of the first apparatus.
3. The request processing system of claim 1, wherein the public key data is generated by using a hash function to convert the public key, and its size is smaller than the original public key.
4. The request processing system of claim 1, wherein the public key data is a public key address used in a blockchain system.
5. The request processing system of claim 1, wherein the first apparatus-identifying specific data is written in a smart contract address of Ethereum.
6. The request processing system of claim 1, further comprising a unit for authenticating a writing

authority of the first apparatus.

7. The request processing system of claim 1, wherein the first apparatus-identifying specific data is stored in a system for collecting a charged fee from a user according to a data size to be stored.

8. The request processing system of claim 1, wherein the unlocking cryptographic data enable to uniquely identify the unlocking cryptographic information includes a unique identifier of a signature scheme and a unique identifier of a parameter of a signature scheme.

9. A request processing method performed at a request processing system, having a first apparatus and a second apparatus associated with the first apparatus for receiving a request sent from the first apparatus at the second apparatus and executing the request, the method comprising: the steps of, executed by a computer being the first apparatus: maintaining a specific secret key and a public key; generating first apparatus-identifying specific data, including public key data enable to uniquely identify the public key and unlocking cryptographic data enable to uniquely identify unlocking cryptographic information; writing the generated first apparatus-identifying specific data into a predetermined address on a communication network; and sending a request with a signature, signed with the secret key, and the public key to the second apparatus, and the steps of, executed by a computer being the second apparatus: reading the first apparatus-identifying specific data written at the predetermined address; authenticating the first apparatus by generating, from the public key received from the first apparatus, public key data enable to uniquely identify the public key; and determining whether this public key is included in the downloaded first apparatus-identifying specific data as well as, based on the determination, validating a signature of the request with a signature based on the public key received from the first apparatus and the unlocking cryptographic data included in the downloaded first apparatus-identifying specific data; and executing the validated request based on the authentication of the first apparatus.

10. The request processing method of claim 9, wherein the first apparatus is a smart key, and the second apparatus is mounted on a vehicle and is a vehicle onboard apparatus for controlling unlocking and locking of the vehicle by the request of the first apparatus.

11. The request processing method of claim 9, wherein the public key data is generated by using a hash function to convert the public key, and its size is smaller than the original public key.

12. The request processing method of claim 9, wherein the public key data is a public key address used in a blockchain system.

13. The request processing method of claim 9, wherein the first apparatus-identifying specific data is written in a smart contract address of Ethereum.

14. The request processing method of claim 9, further comprising the step of authenticating a writing authority of the first apparatus.

15. The request processing method of claim 9, wherein the first apparatus-identifying specific data is stored in a system for collecting a charged fee from a user according to a data size to be stored.

16. The request processing method of claim 9, wherein the unlocking cryptographic data enable to uniquely identify the unlocking cryptographic information includes a unique identifier of a signature scheme and a unique identifier of a parameter of a signature scheme.